

Deference and Infinite Frames^{*}

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Abstract: Three recent results about probabilistic deference, due to Zhang, Geanakoplos, and Dorst et al., each hold for all finite probability frames but fail when frames are allowed to be infinite. Zhang's result, that a novice cannot defer to two experts while planning to always have a credence strictly between them when they disagree, requires a finite range of possible expert credences; a counterexample using normal distributions shows it fails otherwise. Geanakoplos's result, that more informative experiments are more valuable when experiments are reflexive, transitive, and nested, does not extend to uncountable frames with discontinuous payoffs, nor does it extend when both the state space and the option set are infinite. The equivalence Dorst et al. establish between Total Trust and Value breaks down in both directions on infinite frames: Total Trust can hold without Value (when utilities are unbounded), and Value can hold without Total Trust (when the option set is finite but the state space is countably infinite). These failures raise questions about the philosophical significance of results that hold only in the finite case, though the paper largely sets those questions aside in favour of establishing the formal results.

RECENTLY there has been some philosophical interest in the epistemology of *deference*. Some of the important questions about deference concern pooling.¹ When we regard two other people as experts, but they disagree, how should we come up with a view that balances the two views? Some of the questions concern who is an expert. We've known since at least the work of David Blackwell (1953) that given negative and positive introspection for evidence, it is never a bad idea to regard one's more informed self as an expert. In general, however, negative introspection for evidence isn't very plausible, and it would be good to know how much it can be weakened while retaining something like Blackwell's results.

This note concerns three recent (or at least recently published) results relevant to these questions. All of the results hold for arbitrarily large finite frames. Somewhat

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¹See Elkin and Pettigrew (2025) for an up to date account of issues about pooling.

surprisingly, none of them hold in general for infinite frames. I don't have any theory about why this topic should yield so many results that have the interesting characteristic of holding on all finite frames, but not all infinite frames. I also don't have sharp bounds on just when these results do and don't hold. The examples below differ both in the cardinality of the frames used (one is countable, two are uncountable), and the continuity of the value functions defined on them (one makes essential use of a discontinuous value function). So there are plenty of open questions here, although I hope there's still some interest in these results.

1 Dual Deference

The models we'll consider in this paper primarily consist of a novice and some experts. We'll use C for the novice's probability function, and A for the expert's. Note that although A and C both denote probability functions, they do so in different ways. C is a name for the novice's probability function, while A is a description of the expert's unknown function. So we can sensibly talk about the probability of $A(p) = a$, meaning the probability that the expert assigns probability a to p . The expert need not literally be another person; we'll also be interested in the case where 'expert' is the state the novice will be in after conducting an experiment or learning some information.

The strongest way that a novice can defer to an expert (with respect to p) is to take the expert to settle what the probability of p is. Formally, it is that $\forall a: C(p \mid A(p) = a) = a$. Our first question is when C can defer in this strong sense to two different experts A and B .²

There are two cases when this can happen quite easily. The first is when C is certain that A and B will agree, i.e., $C(A = B) = 1$. The second is when C takes one or other of the functions to be superior, i.e., when they disagree to always go with what one particular function says. So if C takes A to be superior, then $\forall a, b: C(p \mid A(p) = a \wedge B(p) = b) = a$. But is there a third option? Can C think that A and B are both worthy of total deference, that they might disagree, and when they do the right thing to do is to land somewhere between their two credences?

Dmitri Gallow (2018) proved one important negative result here. He showed that there is no triple of probability functions C, A, B satisfying the following constraints.

1. $\forall a: C(p \mid A(p) = a) = a$;
2. $\forall b: C(p \mid B(p) = b) = b$;
3. $C(A = B) < 1$;
4. For some $\lambda \in (0, 1)$, $\forall a, b: C(p \mid A(p) = a \wedge B(p) = b) = \lambda a + (1 - \lambda)b$.

²As with A, B here is a description for the second expert's probabilities.

That is, C can't defer to both A and B individually, think that A and B might disagree, and in the event they do disagree, plan to take a fixed linear mixture of A's probability and B's probability as the probability of p . This result, unlike most I'll discuss in this paper, does not make any finiteness assumptions, but it does make a strong assumption about mixing, namely that the mixture will be linear.

Snow Zhang (Forthcoming) recently proved a result that considerably generalises Gallow's result, though it does add one crucial extra condition.³ She shows that it is impossible for A, B and C to satisfy the following five constraints.

1. $\forall a: C(p \mid A(p) = a) = a$;
2. $\forall b: C(p \mid B(p) = b) = b$;
3. $C(A = B) < 1$;
4. For any a, b : $C(p \mid A(p) = a \wedge B(p) = b)$ is strictly between a and b .
5. For some finite set of values S , $C(A(p) \in S \wedge B(p) \in S) = 1$.

This section shows that the last constraint is essential; it is possible to satisfy the first four constraints without it. I'll show this by constructing a model where the first four constraints are satisfied. In this model there will uncountably many values that $A(p)$ and $B(p)$ could take.

Let X , Y and Z be independent normal distributions with mean 0 and variance 1. In symbols, each of them is $\mathcal{N}(0,1)$. So the sum of any two of them has distribution $\mathcal{N}(0,2)$, and the sum of all three has distribution $\mathcal{N}(0,3)$. Let p be the proposition that this sum, $X + Y + Z$, is positive. Let C be a probability function that incorporates all these facts, but has no other direct information about X , Y , and Z . So $C(p) = 1/2$, since in all respects C 's opinions are symmetric around 0.

C knows some things about A and B . Both of them know everything C knows about X , Y , Z , and each are logically and mathematically omniscient, and know precisely what evidence they have.⁴ One of them knows the value of X , and one of them knows the value of $X + Y$. A fair coin was flipped. If it landed heads, then A knows X and B knows $X + Y$; if it landed tails, it was the other way around. C knows about this arrangement, but doesn't know how the coin landed. Let H be the proposition that it landed heads.

Since both A and B know everything C knows plus something more, and satisfy positive and negative introspection, C should defer to them. If C knew which knew $X + Y$ and which only knew X , they would defer to the one who knew $X + Y$. They

³Zhang's result is reported in Pettigrew and Weisberg (2024). The full result Zhang proves is much more general than this one, but the version I'll discuss makes it easier to see the contrast between the finite and infinite case.

⁴That is, each of them satisfy positive and negative introspection for evidence. The next two sections will drop the assumption that more informed functions satisfy negative introspection.

don't know this, but conditional on knowing the values of $A(p)$ and $B(p)$, they can go close to figuring it out.

Assume for now that the coin landed heads, so H is true. We'll work out the joint density function for A and B . Then we can work out the same density function conditional on $\neg H$, and from those two facts work out the posterior probability of H . Call this value b . Conditional on $A(p) = a$, and $B(p) = b$, C 's probability for p should be $(1-b)a + bb$. That's because conditional on $A(p) = a$, $B(p) = b$ and H , C 's probability for p should be b , while conditional on $A(p) = a$, $B(p) = b$ and $\neg H$, C 's probability for p should be a . The short version of what follows is that since b is a function of a and b and is always in $(0,1)$, it follows that C obeys constraint 4.

Given H , we can work out the value of X from $A(p) = a$. In what follows, $\Phi(x)$ is the cumulative distribution for the standard normal distribution, i.e., for $\mathcal{N}(0,1)$, and Φ^{-1} is its inverse. If $X = x$, then p is true iff $Y + Z > -x$. Since $Y + Z$ is a normal distribution with mean 0 and variance 2, i.e., standard deviation $\sqrt{2}$, the probability of this is $\Phi(\frac{x}{\sqrt{2}})$. So $x = \sqrt{2}\Phi^{-1}(a)$.

Given H , that $X = \sqrt{2}\Phi^{-1}(a)$, and $B(p)$, we can work out what Y must be as well. If $B(p) = b$, that means that the probability that $Z > -(X + Y)$ is b . Since Z just is a standard normal distribution, that means that $X + Y$ is $\Phi^{-1}(b)$, and hence Y is $\Phi^{-1}(b) - \sqrt{2}\Phi^{-1}(a)$.

Now we can work out the joint density function for a and b conditional on H . Given H , $A(p) = a$ and $B(p) = b$ just when $X = \sqrt{2}\Phi^{-1}(a)$ and $Y = \Phi^{-1}(b) - \sqrt{2}\Phi^{-1}(a)$. And if we write $\phi(x)$ for the density function for the standard normal distribution⁵, the joint distribution for $A(p) = a \wedge B(p) = b$ given H has density

$$\phi(\sqrt{2}\Phi^{-1}(a))\phi(\Phi^{-1}(b) - \sqrt{2}\Phi^{-1}(a))$$

By a parallel calculation, the joint density function for $A(p) = a \wedge B(p) = b$ given $\neg H$ has density

$$\phi(\sqrt{2}\Phi^{-1}(b))\phi(\Phi^{-1}(a) - \sqrt{2}\Phi^{-1}(b))$$

So given that $A(p) = a \wedge B(p) = b$, the probability of H is

$$\frac{\phi(\sqrt{2}\Phi^{-1}(a))\phi(\Phi^{-1}(b) - \sqrt{2}\Phi^{-1}(a))}{\phi(\sqrt{2}\Phi^{-1}(a))\phi(\Phi^{-1}(b) - \sqrt{2}\Phi^{-1}(a)) + \phi(\sqrt{2}\Phi^{-1}(b))\phi(\Phi^{-1}(a) - \sqrt{2}\Phi^{-1}(b))}$$

If we call that value λ , it follows that $C(p \mid A(p) = a \wedge B(p) = b) = \lambda b + (1-\lambda)a$, and since $\lambda \in (0,1)$, this means that C satisfies constraint 4. This is consistent with Gallow's

⁵i.e., $\phi(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}$.

result because λ is not a constant, it is a function of a and b . And it is consistent with Zhang's result because each of $A(p)$ and $B(p)$ can take infinitely many, in fact uncountably many, values. If one tries to make a similar construction to this one with only finitely many possible values for the probabilities, there will be some value which only the more informed probability can take, and in that case C 's posterior probability will be equal to the probability of the more informed expert.

To understand the relationship between a , b , and C 's posterior probability, it helps to visualise one part of it. Figure 1 shows what value this posterior takes for different values of b holding fixed $a = 0.75$.

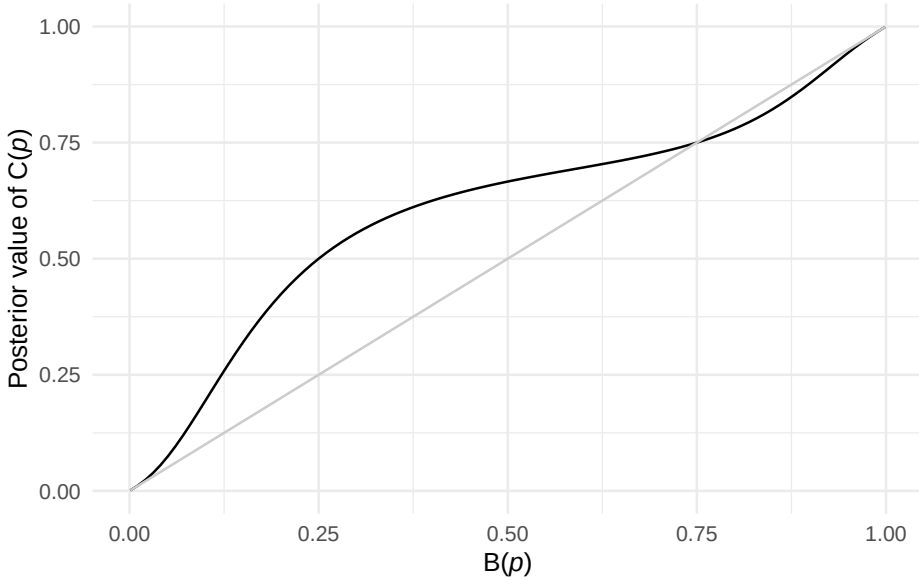


Figure 1: The posterior probability of $C(p)$ given $A(p) = 0.75$.

The distribution loosely follows what Levinstein (2015) calls Thrasymachus's Principle. The more opinionated of the two experts gets much stronger weight. You can see this in part by seeing how close the above graph gets to $x = y$ at either extreme. But it's perhaps more vivid if we plot the posterior probability that the coin landed Tails against the different values of $B(p)$, as in Figure 2.

When $B(p)$ is between 0.25 and 0.75, i.e., when it is closer to 0.5 than $A(p)$ is, C is confident that the coin landed Tails, and that A is more informed and hence more worthy of deference. When $B(p)$ takes a more extreme value, then C is confident that the coin landed Heads, and hence that B is more worthy of deference. In general, this model backs up Levinstein's intuition that more opinionated sources are probably better informed, and hence more worthy of deference.

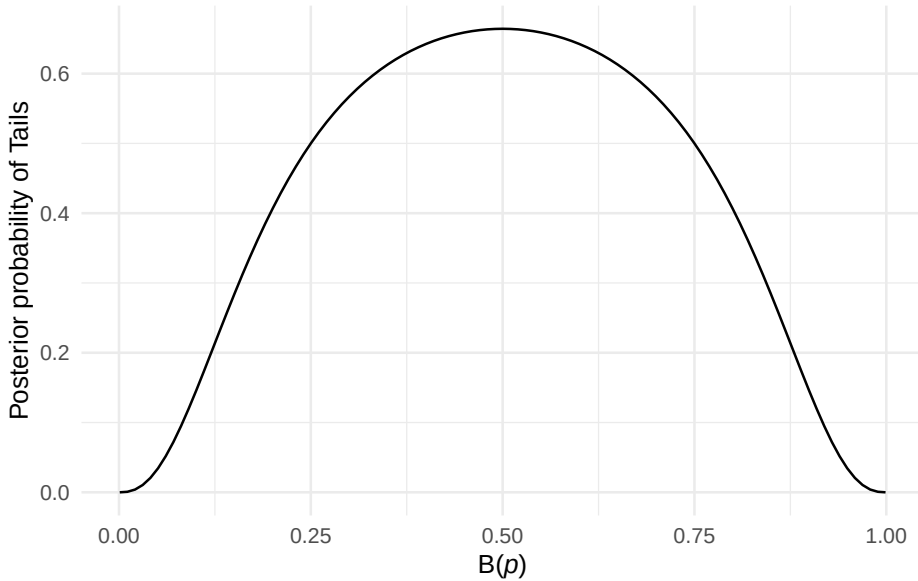


Figure 2: The posterior probability of the coin landing Tails given $A(p) = 0.75$.

2 Evidence and Nesting

The previous section assumed that C strongly deferred to A and B. I now turn to the question of when C should do that. A natural thought, one I relied on in that discussion, was that C should defer when they regard A and B as better informed than they are. This can be motivated with a famous result from David Blackwell (1953). Let E_1 and E_2 be functions from W to subsets of W . Intuitively, these are *experiments*; The Experimenter will perform E_i and learn they are in $E_i(w)$, where w is the world they are in. They will then update by conditionalising⁶ on $E_i(w)$. Blackwell assumes that the range of each E_i is a partition of W ; The Experimenter always learns what cell of the partition they are in.

The short version of the big result is that E_1 is guaranteed to be more valuable than E_2 iff E_1 is more informative than E_2 . All of that needs clarifying though.

Say E_1 is a *refinement* of E_2 iff for all w , $E_1(w) \subseteq E_2(w)$. Formally, this is how I'll capture the intuitive notion of being more informative.⁷

⁶This is a substantive and controversial assumption. There has been some dispute about whether conditionalisation is the correct rule of update if evidence doesn't satisfy positive or, as in this case, negative introspection. I'm simply going to assume here that conditionalisation is the correct updating rule even if negative introspection fails, and note that a lot of things would change if that were not correct. For more on this debate, see Miriam Schoenfeld (2017), Dmitri Gallow (2021), Yoav Isaacs and Jeffrey Sanford Russell (2023), Pablo Zendejas Medina (2023), and Ru Ye (forthcoming).

⁷Note that comparisons here always include equality. An experiment is a refinement of itself, is more in-

Let O be a finite set of options: $\{O_1, \dots, O_n\}$. Each O_i is a function from W to reals. Intuitively, they are bets, and the number is the return on each bet. I'll follow standard terminology in philosophy and say that a function from worlds to reals is a *random variable*. Given a random variable X (defined on W) and a probability function Pr , we can define the expectation $Exp(X, Pr)$ as $\sum Pr(w)X(w)$, where the sum is across members of W .⁸

Say a strategy S is a function from E and W to O such that if $E(x) = E(y)$, then $S(x) = S(y)$. That is, strategies are not more fine-grained than evidence. Intuitively, a strategy is something that The Experimenter can implement given their evidence, so it can't require them to make more discriminations than their evidence does. For each S , we can define a random variable S_R (read this as the return of S), such that $S_R(w) = S(w)(w)$. In words, the return of S at w is the value at w of the option S selects at w .

Finally, say that a strategy is **recommended**⁹ by Pr (relative to E , O and W) just in case for all w in W , and alternative options O_a in O , $Exp(S(w), Pr(\bullet | E(w))) \geq Exp(O_a, Pr(\bullet | E(w)))$. In words, the option selected by S at w has maximal expected utility out of the options in O , relative to the result of updating Pr on the evidence available at w . Given these notions, we can state two important results Blackwell proves.

First, for any O , W and Pr , if E_1 is a refinement of E_2 , S_1 is recommended by E_1 and S_2 is recommended by E_2 , then $Exp(S_1, Pr) \geq Exp(S_2, Pr)$. No matter what practical problem The Experimenter is facing, and no matter what their priors are, they are better off adopting a strategy recommended by the more informative experiment.

Second, for any W , if E_1 is not a refinement of E_2 , then for some O and Pr , there exists an S_1 is recommended by E_1 and S_2 is recommended by E_2 such that $Exp(S_1, Pr) < Exp(S_2, Pr)$. That is, if E_1 is not more informative than E_2 , then for some practical problem, it is better in expectation to perform E_2 and carry out some strategy recommended by it.

Blackwell isn't the first to connect the value of experiments to the relative value of strategies and options in this way; for some history of this idea see Das (2023) and Cam (1996). But he works out the consequences of it in much more detail than anyone

formative than itself, and is more valuable than itself. This can lead to confusion, but it's the standard terminology, and the alternative is much more wordy.

⁸This definition of expected value needs to be generalised in familiar ways if W is uncountable. I'm defining random variable and expected value the way they are usually defined in philosophy. In some fields it is more common to define random variables as functions from a probability space to reals, where a probability space has W and Pr as constituents. Then we can define expectation as a one-place function that simply takes a random variable as input. I think the philosophers' way of speaking is more useful, and in any case I'm a philosopher so it's more natural to me. But note there is a potential terminological confusion here.

⁹This term is taken from Dorst et al. (2021)

had before him. (The two results I've listed here don't go close to exhausting what he proved, but they are enough for our purposes.) Philosophers have primarily focussed on the first of these two results. And they have focussed largely on the special case when $E_2(w) = W$; i.e., when 'performing' experiment E_2 means getting no information at all.¹⁰ In this section we'll also ignore the second result, but we will pay attention to the case where E_2 isn't universal.

John Geanakoplos ([1989] 2021) proved an interesting generalisation of this first result. As noted earlier, Blackwell's theorems presuppose that each experiment is partitional. Formally, E is partitional iff for all w , $w \in E(w)$, and for all w, v , if $v \in E(w)$, then $E(w) = E(v)$. Geanakoplos shows something interesting about experiments that are reflexive, transitive, and nested. These are defined as follow (with leading quantifiers over worlds left implicit)

Reflexive $w \in E(w)$.

Transitive If $v \in E(w)$, then $E(w) \subseteq E(v)$.

Nested Either $E(w) \cap E(v) = \emptyset$, or $E(w) \subseteq E(v)$, or $E(v) \subseteq E(w)$.

He shows that both of Blackwell's results continue to hold when E_1 is reflexive, transitive, and nested, as long as E_2 is partitional.

This result has had some influence on recent philosophical work. It suggests the following kind of argument.

1. Performing experiments is valuable.
2. Performing experiments is valuable iff experiments are nested.
3. Therefore, experiments are nested.

That's fairly crude as stated, but it's possible to develop it into a more sophisticated argument that has implications for what the correct epistemic logic should be. You can find (more sophisticated) versions of this argument in Spencer (2018) and Dorst (2019), and criticisms of these arguments in Williamson (2019) and Das (2023).

The aim of this section is to show that two of the assumptions that Geanakoplos uses in proving these results are essential. First, E_2 has to be partitional; it is not sufficient that it is reflexive, transitive, and nested. Second, it cannot be that both W and O are infinite. Both results arguably help the Williamson-Das side of the debate mentioned in the previous paragraph, but I won't go into that in more detail here. The aim is just to show the formal limits of Geanakoplos's result.

¹⁰Further, they haven't always credited Blackwell (or the earlier results from Peirce and Ramsey that Das discusses) when they do discuss the results. I'm grateful to John Quiggin for pointing out to me the importance of Blackwell's results in this context.

Here is the model that shows that more refined experiments do not necessarily have higher expected value if both experiments are reflexive, transitive, and nested.

- $W = \{w_1, w_2, w_3\}$
- $\Pr(w_1) = \Pr(w_2) = \Pr(w_3) = \frac{1}{3}$
- $O = \{O_1, O_2\}$
- $O_1(w_1) = O_1(w_2) = O_1(w_3) = 0$
- $O_2(w_1) = 3$
- $O_2(w_2) = 9$
- $O_2(w_3) = -6$
- $E_1(w_1) = \{w_1, w_3\}$
- $E_1(w_2) = \{w_2\}$
- $E_1(w_3) = \{w_3\}$
- $E_2(w_1) = \{w_1, w_2, w_3\}$
- $E_2(w_2) = \{w_2, w_3\}$
- $E_2(w_3) = \{w_3\}$

Given no information, the optimal strategy is to always take the bet, i.e., choose O_2 over the fixed return of 0 that is O_1 . This has an expected return of 2. Given E_1 , the only recommended strategy is to choose O_1 at w_1 and w_3 , and O_2 at w_2 , for an expected return of 3. But given the less informative E_2 , the recommended strategy is to choose O_2 at w_1 and w_2 , and O_1 at w_3 , for an expected return of 4. In this case, performing the less informative experiment has higher expected returns. (Though to be clear, both experiments have positive expected returns, relative to not doing anything.)

Next I'll show the result does not hold when W and O are infinite. It pays to be careful here because it's easy to have a case where S_1 and S_2 are undefined. And it's not interesting that $\text{Exp}(S_1, \Pr) \geq \text{Exp}(S_2, \Pr)$ might sometimes fail to be true simply because one or other term in it isn't defined. So I'll restrict attention to cases where utilities are bounded, for any E and any w there is an optimal strategy given $E(w)$, and the expectation of any recommended strategy is defined.

Let W be the reals in $(0,1)$. $E_1(x) = [x,1)$, and $E_2(x) = (0,1)$. For any x in $[0,1]$, let $O_x(y)$ be -1 if $x \leq y$, and x if $x > y$, and O be the set of all these O_x . And $\Pr$ is the flat distribution over $(0,1)$; the only fact I'll need is that whenever $0 < x < y < 1$, the probability that the actual world is in (x, y) is $y - x$.

The only strategy recommended by E_2 is to always choose O_0 , which has a guaranteed return of 0. The strategy recommended by E_1 is to choose O_x upon learning that the true value is in $(x, 1)$. That has an expected return of x , which is higher than all the alternatives. But following that strategy all the time has a guaranteed return of -1 ,

which is worse than the strategy recommended by E_2 . And that's true even though E_2 is partitional, and E_1 is reflexive, transitive, and nested.

Now there is something odd about this example - each of the O_x is discontinuous. In each case, the payouts jump from -1 to x at a particular point. This suggests a further open question: If every member of O is continuous, are more refined experiments valuable in expectation? It's also, as far as I know, open whether Geanakoplos's results hold in countable frames. But what this model shows is that his result does not generalise to uncountable frames with discontinuous payouts.

3 Trust and Value

The role of experiments in the frames discussed in Section 2 is somewhat curious. They are in one respect central, the theorems are all restricted to whether frames are partitional, nested, etc, but they are in another respect ephemeral. Ultimately what matters is not the experiment, but the probability that The Experimenter has after performing an experiment. This latter way of thinking is a helpful way to understand a striking recent result by Dorst et al. (2021).

Say a probability frame is an ordered pair $\langle W, P \rangle$ such that W is a set (intuitively, of worlds), and P is a function from W to probability functions defined on w . One way to generate such a pair is to have some experiment E and prior probability Pr , each defined on W , and have $P(w)$ be $Pr(\bullet \mid E(w))$. But you can just cut out the E and Pr , and focus simply on W and P . As Dorst et al. (2021) show, this turns out to be mathematically a very helpful move. It lets you see a lot more interesting features of these frames.¹¹ They call frames where P is generated from E and Pr in this way *prior frames*, and those will be the focus of discussion here, but it is interesting to see them as a special case of a more general class.

One thing they prove by looking at the more general class is that when W is finite, the following two claims are equivalent. (I'll state the claims formally, then explain my notation.)

Total Trust $E(X \mid \{w: \text{Exp}(X, P(w)) \geq t\}, \pi) \geq t$

Value If O is a set of options, s is a recommended strategy for O , and o is a member of O , then $\text{Exp}(s, \pi) \geq E(o, \pi)$.

There is a bit there to unpack. I'll follow them in using π for a probability function that is outside the frame. I'll sometimes call it Novice's probability function, as opposed to

¹¹At least in the case where W is finite; I'll be getting to the issues when it is not.

$P(w)$ which is The Experimenter's probability function at w .¹² I'm generalising the notion of expectation a bit to allow for conditional expectations; $\text{Exp}(X \mid p, \text{Pr})$ is the expectation of X according to $\text{Pr}(\bullet \mid p)$. So here's what Total Trust says. Take any random variable X . Update π by conditionalising on the proposition that consists of all and only worlds where The Experimenter at that world has an expected value for X at least equal to t . After that update, Novice also expects X 's value to be at least t .

I discussed recommended strategies in Section 2, so there is less to say about Value. What it says is that Novice does not expect any member of O to do better than any recommended strategy.

One way Dorst et al put the equivalence between Total Trust and Value (on finite frames) is that π Totally Trusts a frame $\langle W, P \rangle$ iff it Values that frame. What I'll show is that this equivalence breaks down without the finiteness assumptions. Indeed, it breaks down even when W is countably infinite.

Start with a frame I'll call **Coin**. A fair coin will be flipped repeatedly until it lands Tails. Let F be a random variable such that $F = x$ iff the coin is flipped x times. (If the coin never lands Tails, I'll stipulate that $F = 1$. Since this has probability 0, it doesn't make a difference to the probabilities, but it will make a difference to the possible choices post-update.) Novice knows these facts about F , so $\pi(F = x) = 2^{-x}$. If $F = x$, then The Experimenter learns $F \geq x$ and nothing else, and updates on that. That is, $P(F = x) = \pi(\bullet \mid F \geq x)$. For any positive integer i , let O_i be the random variable that takes value 0 at $F = j$ when $j \leq i$, and value 2^i at $F = j$ when $j > i$. Let O be the set of each O_i . The strategy s such that $s(F = i) = O_i$ is recommended, as can be easily checked. But $E(s) = 0$, while for any $o \in O$, $E(o, \pi) = 1/2$. So Value fails on **Coin**.

On the other hand, π does Totally Trust **Coin**. For any random variable X and threshold t , say an integer k is a cut-off if either $\text{Exp}(X \mid F \geq k, \pi) \geq t$ and $\text{Exp}(X \mid F \geq k + 1, \pi) < t$, or $\text{Exp}(X \mid F \geq k, \pi) < t$ and $\text{Exp}(X \mid F \geq k + 1, \pi) \geq t$. Let c_i be the i 'th cutoff. Partition the integers into the regions between cutoffs. More precisely, do the following. If 1 is not a cutoff, the first cell of the partition is $\{1, \dots, c_1 - 1\}$; otherwise the first cell is just $\{1\}$. If there is a last cutoff c , the last cell is $\{c, c + 1, \dots\}$. Otherwise, each cell is $\{c_i, \dots, c_{i+1} - 1\}$. Say a cell is *positive* if for every k in it, $\text{Exp}(X \mid F \geq k, \pi) \geq t$, and negative otherwise. (By the construction of the cells, $\text{Exp}(X \mid F \geq k, \pi) \geq t$ is true for either all or none of the members.)

Let $\{c_i, \dots, c_{i+1} - 1\}$ be an arbitrary positive cell. By construction, $\text{Exp}(X \mid F \geq c_i, \pi) \geq t$, and $\text{Exp}(X \mid F \geq c_{i+1}, \pi) < t$. Since for some $\lambda \in (0, 1)$, $\text{Exp}(X \mid F \geq c_i, \pi) = \lambda \text{Exp}(X \mid F \in \{c_i, \dots, c_{i+1} - 1\}, \pi) + (1 - \lambda) \text{Exp}(X \mid F \geq c_{i+1}, \pi)$, it follows that $\text{Exp}(X \mid F \in \{c_i, \dots, c_{i+1} - 1\}, \pi) \geq t$. Since this was an arbitrary positive cell, it follows that $\text{Exp}(X \mid F \in I, \pi) \geq t$

¹²When the frame is a prior frame, it is natural to focus on the case where $\pi = \text{Pr}$, but again I'm not just looking at prior frames.

for any positive cell I . Since $\text{Exp}(X \mid \{w: \text{Exp}(X, P(w)) \geq t\}, \pi)$ is a weighted average of the values of $\text{Exp}(X \mid F \in I, \pi)$ where I is one or other of the positive cells, it follows that $E(X \mid \{w: \text{Exp}(X, P(w)) \geq t\}, \pi) \geq t$, as required.

So π Totally Trusts **Coin**, but doesn't Value it. So the equivalence between Total Trust and Value fails here. But you might very reasonably object on two scores. First, the value function used to generate the counterexample was unbounded, and we know that unbounded value functions lead to all sorts of paradoxes. Second, I didn't just make W infinite, I made O infinite as well, so this isn't a minimal generalisation of the original claim. It turns out that if we put both these constraints on, then the equivalence fails in the other direction: It is possible to get a frame that π Values, but does not Totally Trust.

Call the following frame **Bentham**. Again, a coin will be flipped until it lands Tails. If it ever lands Tails, F is the number of flips. If it never lands Tails, which has probability 0, then $F = \infty$. Again, Novice knows these facts, and so far the case is just like **Coin**. But in this case, if $F = x$, The Experimenter learns that $F \leq x$ and nothing else, and The Experimenter updates on that. So if $F = \infty$, The Experimenter learns nothing, but otherwise they can rule out all but finitely many possibilities. More precisely, $P(F = x) = \pi(\bullet \mid F \leq x)$.

The Novice probability does not Totally Trust this frame. Let Y be a random variable such that $Y(F = \infty) = 0$, and for all finite n , $Y(F = n) = 1 - 2^{-n}$. $E(Y \mid \{w: \text{Exp}(Y, P(w)) \geq \frac{2}{3}\}, \pi) = 0 < \frac{2}{3}$. The only world w where $\text{Exp}(Y, P(w)) \geq \frac{2}{3}$ is $F = \infty$, and at $F = \infty$, $Y = 0$.

On the other hand, π does Value this frame. To see this, for any set of options O , recommended strategy S , random variable X (all defined on W), and integer n , let W_n be the set $\{F = 1, \dots, F = n\}$, O_n, P_n, S_n and X_n be the restrictions of O, P and S to worlds in W_n . From the way P is constructed, i.e., by conditionalising on the set of worlds where F is no greater than it actually is, it follows that if S is recommended on $\langle W, P \rangle$, then S_n is recommended on $\langle W_n, P_n \rangle$. Since $\langle W_n, P_n \rangle$ is a finite prior frame where E is reflexive, transitive and nested, and $\text{Pr} = \pi$, it follows by the result of Geanakoplos described in Section 2, that the expected return of S_n is greater than the expected return of any option in O_n . For any random variable X , $\text{Exp}(X, \pi)$ is the limit as n tends to ∞ of $\text{Exp}(X_n, \pi)$; this is because as n grows this covers all worlds in W except $F = \infty$, which has probability 0. If the expected return of S is the limit n tends to ∞ of the expected return of S_n , and the expected return of an option in O is the limit as n tends to ∞ of its counterpart in O_n , and S_n is better (in expectation) than every option in O_n , it follows that S is better (in expectation) than every option in O . So Value is satisfied, as required.

4 Conclusion

It's striking that we get such different behaviours between finite and infinite frames when it comes to these three somewhat distinct issues about deference and updating. The main point of this note is to point out these differences.

But there is a broader philosophical question. One might think that since humans are finite, results that hold on all finite frames should be used when thinking about humans. I think this is a bit quick. All the models I've been using here, both the finite and the infinite ones, really are *models*. Even the finite ones assume that the people being modeled have superhuman (if not literally infinite) computational capacities. They are all idealisations. The question is, are they good or bad idealisations? Here the issues about finitude get complicated. It might be that an infinite model is a better idealisation, a better approximation to a human than a finite one.

Imagine someone saying that since humans are finite, and circles are infinitely curved, we should never model humans as thinking about circles. Rather, we should think that the human is thinking of a regular polygon with arbitrarily many sides. This is a little absurd. A model where the human is thinking of a circle is simpler than a model where the human is thinking, in full precision, about a chiliagon. By the same reasoning, it might be better to model a scientist as taking some variable to be normally distributed over an interval, than to take them to have in their head a particular finite approximation to the normal distribution, which they perfectly update. For that reason, the model in Section 1 might be a decent model of deference.

The models in Section 2 are, admittedly, weirder. There is less use, even in idealisations, for infinite models with discontinuous payouts, or for models with unbounded utility functions. These are known to lead to weird results. Here I think there is a stronger claim that the models I've presented are not useful models, not because they are infinite, but because they are discontinuous and unbounded.

But there is obviously much more to say about these questions about usefulness. Hopefully it is helpful to simply point out how differently the finite and infinite cases behave.

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